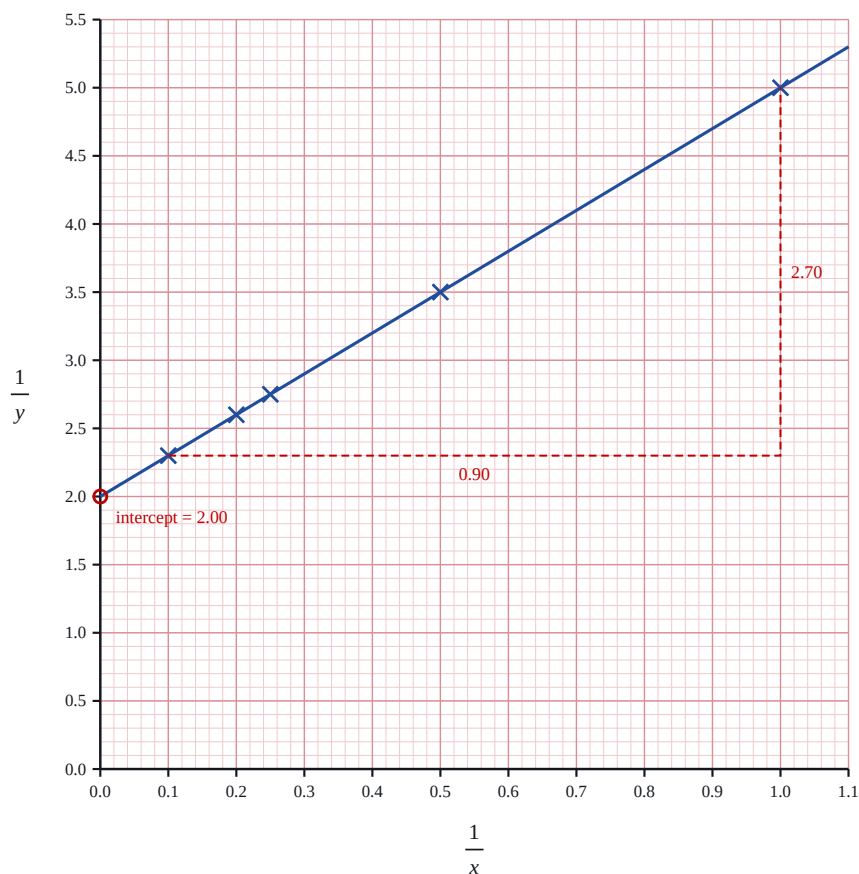


- 9 The variables x and y are related by $y = \frac{x}{ax+b}$, where a and b are constants. (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



Gradient $b = 2.70 \div 0.90 = 3$ · vertical intercept $a = 2.00$ · the ruled line is $1/y = 2 + 3(1/x)$

WORKING — BLUE INK

$$y = \frac{x}{ax+b} \Rightarrow \frac{1}{y} = \frac{ax+b}{x} = a + b\frac{1}{x}$$

Comparing with $Y = mX + c$: plot $\frac{1}{y}$ against $\frac{1}{x}$; gradient b , vertical intercept a .

$1/x$	1.00	0.50	0.25	0.20	0.10
$1/y$	5.00	3.50	2.75	2.60	2.30

MARKS & NOTES — RED INK

Key concept: invert *both* sides first — the linear pair is $1/y$ against $1/x$.

- **B1** takes reciprocals to reach $1/y = a + b(1/x)$
- **B1** states: plot $1/y$ against $1/x$, gradient b , intercept a
- **B1** both reciprocal rows computed, all five points plotted and a ruled straight line drawn through them
- **Trap** plotting y against $1/x$, or $1/y$ against x , gives a curve — no marks for a line of best fit through it
- **Note** the five points are *exactly* collinear here, so a correct plot leaves no scatter; any visible kink means a reciprocal was mis-computed
- **Note** extend the line to cut the vertical axis — part (b) reads the intercept off it, so a line stopped at the first point costs a mark there